

**Maisagip: A Hybrid Deep Learning Framework for Corn Ear Quality Grading and
Defect Detection Using Computer Vision**

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CHAPTER 1

Introduction

Agriculture provides an important role in supporting food security, economic development, and rural livelihoods. With the increasing demand for agricultural products, ensuring the quality and safety of harvested crops has become essential. Computer vision and deep learning have emerged as innovative technologies that provide automated, efficient, and non-destructive methods for agricultural quality inspection. These technologies allow systems to analyze digital images, detect visual defects, and evaluate product characteristics with improved accuracy and consistency compared with traditional manual inspection methods.

This study focuses on the development of Maisagip, a hybrid deep learning framework designed to automate the post-harvest quality grading and defect detection of harvested corn ears. The system integrates image classification, object detection, and rule-based decision support to identify corn varieties, detect visible defects, estimate physical characteristics, determine quality grades, and generate inspection recommendations. By applying artificial intelligence and computer vision techniques, the proposed system aims to provide a standardized and efficient approach to corn quality assessment.

Along with this, an automated computer vision-based inspection system reduces the limitations of manual grading processes, which are often time-consuming, subjective, and inconsistent. Through the use of digital image analysis and deep learning models, agricultural stakeholders may perform faster and more reliable quality evaluations, improve post-harvest decision-making, reduce quality assessment errors, and support better management of harvested corn products. The study also highlights the potential of artificial intelligence in improving

agricultural practices by providing accessible and practical solutions for farmers, cooperatives, traders, and other stakeholders.

Background of the Study

Agriculture remains one of the primary sectors supporting economic development, food security, and rural livelihoods in the Philippines. Among the country's major agricultural commodities, white corn and yellow sweet corn are important crops cultivated primarily for human consumption. According to the Food and Agriculture Organization (FAO), corn is one of the world's most important cereal crops because of its significant contribution to food supply and human nutrition. Maintaining grain quality is essential for ensuring market value, consumer acceptance, and efficient post-harvest handling.

Following harvest, white corn and yellow sweet corn undergo quality inspection before they are marketed, distributed, or processed. This process involves evaluating the physical characteristics of each harvested corn ear, including kernel completeness, ear size, color uniformity, and the presence of visible defects such as insect damage, mold growth, discoloration, deformities, and missing kernels. These observable characteristics are commonly used to determine product quality and market grade because they directly influence the suitability of corn for human consumption, food processing, or other agricultural applications.

In the Philippines, corn quality evaluation follows recognized grading principles established by the Philippine National Standards (PNS/BAFS), which provide standardized criteria for assessing agricultural commodities based on their physical quality characteristics. These standards promote consistency, fairness, and objectivity in quality inspection by defining

acceptable levels of defects and quality requirements for agricultural products intended for commercial distribution. Applying standardized grading criteria helps ensure that harvested corn meets quality expectations while supporting fair pricing and efficient post-harvest management.

Despite being widely practiced, manual quality inspection presents several limitations. Conventional inspection is often time-consuming, subjective, and inconsistent, especially when applied to large-scale agricultural operations. As production demands continue to increase, manual inspection alone is no longer sufficient to provide rapid and accurate quality evaluation, creating the need for intelligent automated inspection systems (Song et al., 2023). These limitations may result in inconsistent grading decisions, reduced operational efficiency, and variations in product quality during post-harvest handling.

Recent advancements in artificial intelligence, particularly in computer vision and deep learning, have provided promising solutions for automating agricultural quality inspection. Computer vision enables machines to analyze digital images and identify relevant visual characteristics without physically damaging agricultural products, while deep learning algorithms automatically learn complex image features that improve recognition accuracy. Compared with conventional image-processing techniques, deep learning models have demonstrated superior performance in agricultural applications, including crop classification, fruit grading, disease detection, defect identification, and grain quality evaluation. Previous studies have demonstrated that deep learning-based approaches outperform traditional image processing methods in identifying product quality and visible defects (Wonggasem et al., 2024).

Several studies have successfully applied deep learning techniques to corn quality inspection. Existing research has demonstrated high performance in detecting maize defects,

classifying kernels, inspecting seed quality, and sorting harvested corn using convolutional neural networks, object detection algorithms, and semantic segmentation models (Chen et al., 2026; Li et al., 2024; Xu et al., 2026; Yang et al., 2024). However, most existing systems perform only a single inspection task, such as defect detection, quality classification, or semantic segmentation (Chen et al., 2026; Li et al., 2024; Xu et al., 2026). Other studies focus primarily on maize kernels or seeds rather than whole harvested corn ears, making them less suitable for practical post-harvest inspection (Li et al., 2024; Yang et al., 2024). Furthermore, some approaches require specialized imaging equipment such as hyperspectral or near-infrared cameras, which increase implementation costs and limit practical adoption among local farmers and agricultural cooperatives (Xu et al., 2026; Yang et al., 2024). Existing systems also rarely provide comprehensive inspection reports or recommendations regarding the appropriate utilization of harvested corn after grading (Chen et al., 2026; Li et al., 2024). These limitations indicate the need for a more practical and integrated quality inspection system capable of performing multiple related tasks simultaneously using standard RGB images.

Another challenge encountered in computer vision-based agricultural inspection is the occurrence of false-positive classifications. Harvested white corn and yellow sweet corn may exhibit considerable variation in appearance due to differences in maturity, quality grade, physical characteristics, visible defects, lighting conditions, viewing angles, and background environments. If these variations are not adequately represented during model training, the system may incorrectly classify or assess the quality of the target corn ears. To improve classification robustness and generalization, the proposed dataset includes diverse image samples of white corn and yellow sweet corn representing different quality grades, levels of visible defects, ear sizes, kernel completeness, and image acquisition conditions. The inclusion of these

diverse classes enables the model to better distinguish variations within the two target corn varieties and reduce false-positive and misclassification errors during actual deployment.

To address these research gaps, this study proposes "Maisagip: A Hybrid Deep Learning Framework for Corn Ear Quality Grading and Defect Detection Using Computer Vision." The proposed system utilizes RGB images of harvested white corn and yellow sweet corn ears and applies image preprocessing techniques before analysis using a hybrid deep learning model. Unlike conventional approaches that perform only a single inspection task, the proposed system simultaneously classifies corn variety, detects visible defects, estimates kernel completeness and ear size, determines the corresponding quality grade based on established grading criteria, generates recommendations regarding the appropriate utilization of the harvested corn (such as human consumption, food processing, animal feed, or rejection), and automatically produces a comprehensive inspection report through a mobile-responsive web application. Integrating these related quality assessment tasks within a unified deep learning framework is expected to improve inspection efficiency while reducing computational redundancy associated with multiple independent models.

This study aims to contribute to the advancement of intelligent agricultural technologies by providing a practical, objective, and accessible computer vision system for post-harvest quality inspection of white corn and yellow sweet corn intended for human consumption. The proposed system is expected to assist farmers, traders, agricultural cooperatives, food processors, and other stakeholders in achieving more consistent grading, reducing manual inspection time, improving post-harvest decision-making, and supporting the adoption of artificial intelligence in Philippine agriculture. Furthermore, the study may serve as a valuable reference for future

researchers exploring computer vision, deep learning, multi-task learning, and intelligent quality assessment systems for agricultural commodities.

Project Objectives

The primary objective of this study is to develop Maisagip, a mobile-responsive web application and a computer vision system that employs a hybrid deep learning framework that provides a mobile view interface for accessibility and for the automated quality grading and defect detection of harvested white corn and yellow sweet corn ear using digital images. The system aims to assist post harvest quality inspection by providing accurate quality assessment and defect identification. Specifically, this study aims to:

1. To develop and prepare an annotated image dataset of harvested white corn and yellow sweet corn with different quality conditions for training, validation, and evaluation of the proposed hybrid deep learning model.
2. To design and develop a hybrid deep learning model that integrates object detection and image classification models for the automated detection of physical defects and quality grading of harvested corn ears.
3. To develop Maisagip, a mobile-responsive web application and a computer vision system that enables users to upload images of harvested corn ears and automatically perform quality grading and defect detection.
4. To integrate an automated inspection and decision- support module that generates quality grades, identifies detected defects, estimates confidence scores, and provides utilization recommendations based on established corn grading standards.

5. To evaluate the performance of the developed system using object detection and image classification metrics, including mean Average Precision, precision, recall, F1-score, Intersection over Union (IoU), accuracy and inference time, to determine its effectiveness and reliability for automated post-harvest quality inspection.

Significance of the Study

This study contributes to the advancement of intelligent agricultural technologies by developing Maisagip, a mobile-responsive web application and a computer vision system that utilizes hybrid deep learning models for the automated quality grading and detection of harvested white corn and yellow sweet corn ears.. By integrating deep learning–based defect detection, automated quality grading, inspection reporting, and decision-support recommendations into a unified platform, the proposed system aims to provide an objective, accurate, efficient, and standardized approach to post-harvest corn quality inspection. Furthermore, the study seeks to demonstrate the practical application of artificial intelligence and computer vision in improving post-harvest quality assessment, reducing the limitations associated with manual inspection, and supporting data-driven decision-making in the agricultural sector. Moreover, this study could be of importance to the following:

Department of Agriculture

The proposed system may support the Department of Agriculture in promoting the adoption of artificial intelligence and digital technologies for improving agricultural quality management and post-harvest practices. Through automated quality grading, defect detection, and data-driven inspection reports, Maisagip may assist in monitoring corn quality, establishing

more consistent assessment procedures, and supporting evidence-based decision-making for agricultural programs and initiatives. The system may also contribute to future efforts in modernizing agricultural inspection processes and improving the efficiency of quality evaluation for locally produced white corn and yellow sweet corn.

Farmers

The proposed system may assist farmers in evaluating the quality of harvested white corn and yellow sweet corn before marketing, storage, or distribution. Through automated quality grading, defect detection, and utilization recommendations, the system may help farmers make informed post-harvest decisions, reduce grading inconsistencies, minimize post-harvest losses, and improve the market value of their produce.

Consumers

The proposed system may benefit consumers by promoting access to safer and higher-quality corn products through more consistent and objective quality assessment. By identifying visible defects and determining the suitability of harvested corn for human consumption, food processing, or other applications, the system may help improve transparency and confidence in corn quality. The implementation of automated inspection may also support the availability of agricultural products that meet established quality standards, contributing to improved consumer satisfaction and food quality assurance.

Agricultural Cooperatives and the Agricultural Industry

Agricultural cooperatives and stakeholders within the agricultural industry may utilize the system to establish a standardized post-harvest quality inspection process for harvested white

corn and yellow sweet corn. The automated grading, defect detection, and inspection reports may improve quality monitoring, facilitate the collection, storage, processing, and distribution of harvested corn, and promote consistency in quality assessment across the supply chain. By providing objective and efficient quality evaluation, the system may enhance post-harvest management, support quality assurance, and contribute to improved operational efficiency and market competitiveness within the agricultural sector.

Traders and Retailers

The system may assist traders, retailers, and buyers in conducting faster and more consistent quality inspections during procurement and distribution. Standardized grading results and inspection reports may support fair pricing, improve purchasing decisions, and reduce inconsistencies associated with manual quality evaluation.

Food Processors

Food processing companies may use the inspection results and system-generated recommendations to determine the most appropriate utilization of harvested corn, whether for human consumption, food processing, or animal feed. This may improve raw material selection, maintain product quality, and reduce processing losses caused by poor-quality agricultural inputs.

Researchers and Software Developers

The study may serve as a valuable reference for researchers, and software developers in the fields of Computer Science, and other related disciplines. It demonstrates the practical application of computer vision and multi-task deep learning in solving real-world agricultural problems and may be used for classroom instruction, academic discussions, laboratory activities,

and future research. Furthermore, the proposed framework may guide software developers in designing and developing intelligent computer vision applications that integrate image classification, defect detection, quality assessment, recommendation generation, and automated reporting within a unified web-based system for agriculture and other real-world domains.

Future Researchers

Future researchers may use this study as a foundation for further research on computer vision, deep learning, intelligent agricultural systems, and automated quality inspection. They may also expand the study by using larger datasets, integrating IoT and mobile technologies, and applying more advanced AI models to improve system performance and accuracy.

Scope and Study Delimitations

Scope

This study focuses on the design, development, and evaluation of Maisagip, a mobile-responsive web application, allowing users to access the system using different devices and a computer vision system that utilizes a hybrid deep learning model to automate the post-harvest quality inspection of harvested white corn and yellow sweet corn ears intended for human consumption. The system is designed to assist farmers, agricultural cooperatives, traders, retailers, consumers and food processors by providing an objective, efficient, and standardized quality inspection process through digital image analysis.

The proposed system utilizes annotated RGB images of harvested white corn and yellow sweet corn ears collected from farms in Sariaya Quezon and publicly available agricultural

image repositories and locally acquired photographs. To improve classification robustness and generalization, the proposed dataset includes diverse image samples of white corn and yellow sweet corn representing different quality grades, levels of visible defects, ear sizes, kernel completeness, and image acquisition conditions. The inclusion of these diverse classes enables the model to better distinguish variations within the two target corn varieties and reduce false-positive and misclassification errors during actual deployment.

The developed hybrid deep learning model performs the following quality inspection tasks:

- Classifies the captured image as white corn or yellow sweet corn.
- Detects visible defects, including mold growth, insect damage, discoloration, deformities, missing kernels, and other externally observable defects.
- Determines the quality grade of harvested corn based on established grading criteria.
- Estimates relevant physical characteristics, including ear size and kernel completeness.
- Generates recommendations regarding the appropriate utilization of the harvested corn, such as human consumption, food processing, or rejection when severe quality defects are detected.
- Produces an automated inspection report containing the detected corn variety, identified defects, estimated physical characteristics, assigned quality grade, recommendation, confidence score, and inspection date.

The quality grading performed by the system is based on the observable grading principles specified in PNS/BAFS 98:2022 (Sweet Corn - Product Standard - Grading and Classification) and other recognized agricultural grading references applicable to image-based

assessment. Since the proposed system relies solely on RGB image analysis, only externally observable characteristics such as kernel completeness, ear size, color uniformity, insect damage, mold symptoms, discoloration, deformities, and other visible defects are evaluated.

The performance of the proposed hybrid deep learning model is evaluated using standard machine learning and object detection metrics, including Accuracy, Precision, Recall, F1-score, Intersection over Union (IoU), and Mean Average Precision (mAP), to determine its effectiveness, reliability, and suitability for automated post-harvest quality inspection.

Delimitations

The study is limited to the design, development, and evaluation of a web-based computer vision system that automates the post-harvest quality inspection of harvested corn ears intended for human consumption. To define the boundaries of the research and clarify its intended scope, the following limitations are established:

- The study does not evaluate internal quality attributes such as moisture content, sweetness level, nutritional composition, aflatoxin contamination, internal kernel damage, chemical composition, or microbiological properties, as these require laboratory testing and specialized equipment beyond the capabilities of computer vision. Although additional images of other corn varieties and visually similar agricultural objects are included during model training to improve classification performance and reduce false-positive predictions, the system does not assign quality grades to these additional classes.

- The proposed study focuses exclusively on post-harvest quality inspection. It does not include crop disease diagnosis before harvesting, pest monitoring, yield prediction, soil analysis, irrigation management, fertilizer recommendation, weather monitoring, precision farming, or other smart agriculture applications.
- The system accepts RGB images captured using conventional digital cameras or mobile phone cameras. It does not support hyperspectral imaging, thermal imaging, near-infrared imaging, drone-based image acquisition, robotic harvesting, or automated sorting machinery.
- The recommendations generated by the system are based solely on image-based quality assessment and the adopted grading criteria. The system is intended to assist post-harvest quality inspection and decision-making and should not replace official grading, laboratory analysis, or certification conducted by authorized agricultural agencies and qualified agricultural inspectors.
- The performance of the developed hybrid deep learning model depends on the quality, diversity, and representativeness of the collected image dataset. Factors such as poor image resolution, extreme lighting conditions, image blur, severe occlusion, and uncommon physical characteristics of corn ears may affect the accuracy and reliability of the system's predictions.